YEAR 8 MATHS · CHAPTER 2 · MIND MAP

Angles and geometrical figures

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1 Reviewing angles

Complementary angles add to 90°. Supplementary angles add to 180°. Vertically opposite angles are equal.

35° has a complement of **55°** and a supplement of **145°**

Angles on a straight line add to 180°

Angles around a point add to 360°

Acute: under 90°. Right: 90°. Obtuse: 90° to 180°. Reflex: over 180°.

2 Transversal and parallel lines

With parallel lines: corresponding angles are equal (F), alternate angles are equal (Z), co-interior angles add to 180° (C).

If one angle is 70°: corresponding **70°**, alternate **70°**, co-interior **110°**

Watch out: these rules only work when the lines are parallel.

3 Triangles

Angles add to 180°. An exterior angle equals the two opposite interior angles added together.

Angles 40° and 75°: the third is **65°**, the exterior angle is **115°**

Equilateral: all angles 60°

Isosceles: two equal sides, two equal base angles

4 Quadrilaterals

Angles add to 360°.

Angles 80°, 95°, 110°: the fourth is **75°**

Parallelogram: opposite angles equal

Rhombus: all sides equal

Kite: one pair of opposite angles equal

5 Polygons

Angle sum of an n-sided polygon: $S = (n-2) \times 180°$.

Pentagon: $3 \times 180° = 540°$, each angle of a regular one is $108°$

Hexagon: $4 \times 180° = 720°$, each angle of a regular one is $120°$

6 3D solids and Euler's formula

For solids with flat faces: $F + V = E + 2$ (faces, vertices, edges).

Cube: $6 + 8 = 12 + 2$ ✓

Triangular prism: $5 + 6 = 9 + 2$ ✓

Prisms have two matching ends. Pyramids come to a point.

7 3D coordinates

A point is (x, y, z): along, across, then up. The origin is $(0, 0, 0)$.

$(2, 3, 4)$ is 2 along the x-axis, 3 along the y-axis and 4 up the z-axis

A 5 by 3 by 2 box with a corner at the origin has its opposite corner at $(5, 3, 2)$